

June 2020

Category review: Chips

Retail Analytics



Classification: Confidential



Our 17 year history assures best practice in privacy, security and the ethical use of data

We all have a responsibility to use data for good

Privacy

- We have built our business based on privacy by design principles for the past 17 years
- Quantum has strict protocols around the receipt and storage of personal information
- All information is de-identified using an irreversible tokenisation process with no ability to re-identify individuals.

Security

- We are ISO27001 certified - internationally recognised for our ability to uphold best practice standards across information security
- We use 'bank grade' security to store and process our data
- Comply with 200+ security requirements from NAB, Woolworths and other data partners
- All partner data is held in separate restricted environments
- All access to partner data is limited to essential staff only
- Security environment and processes regularly audited by our data partners.

Ethical use of data

Applies to all facets of our work, from the initiatives we take on, the information we use and how our solutions impact individuals, organisations and society.

Quantum believes in using data for progress, with great care and responsibility. As such please respect the confidence nature of this document.

Executive summary

01

Task 1: Data preparation and customer analytics

Creating and interpreting high-level summaries of the data
Finding outliers and removing these (if applicable)
Checking data formats and correcting (if applicable)

You will also want to derive extra features such as pack size and brand name from the data and define metrics of interest to enable you to draw insights on who spends on chips and what drives spends for each customer segment

02

Task 2: Experimentation and uplift testing

Evaluate the performance of a store trial which was performed in stores 77, 86 and 88.

Once you have selected your control stores, compare each trial and control pair during the trial period.

You want to test if total sales are significantly different in the trial period and if so, check if the driver of change is more purchasing customers or more purchases per customers etc

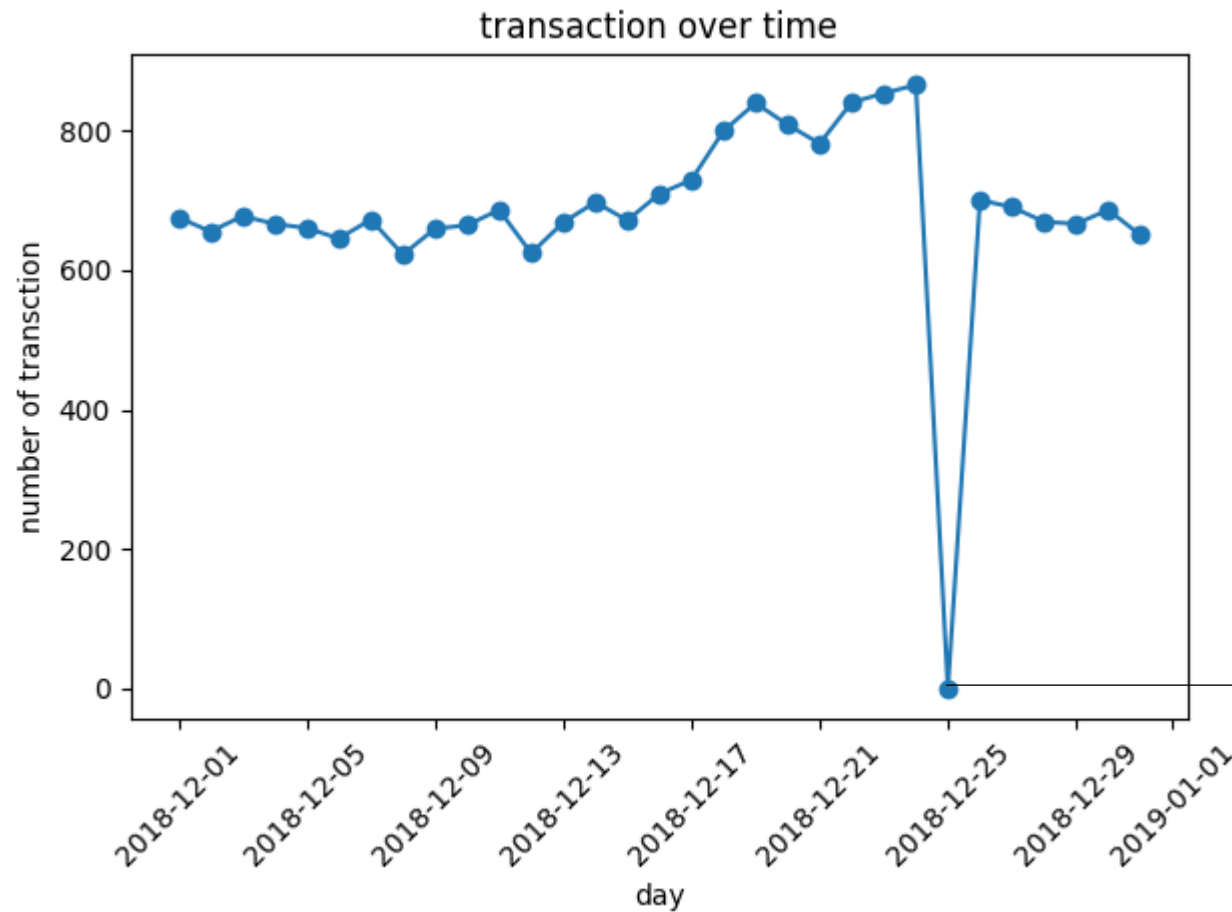
01

Task 1 – Data Preparation & Customer Analytics

Objective:

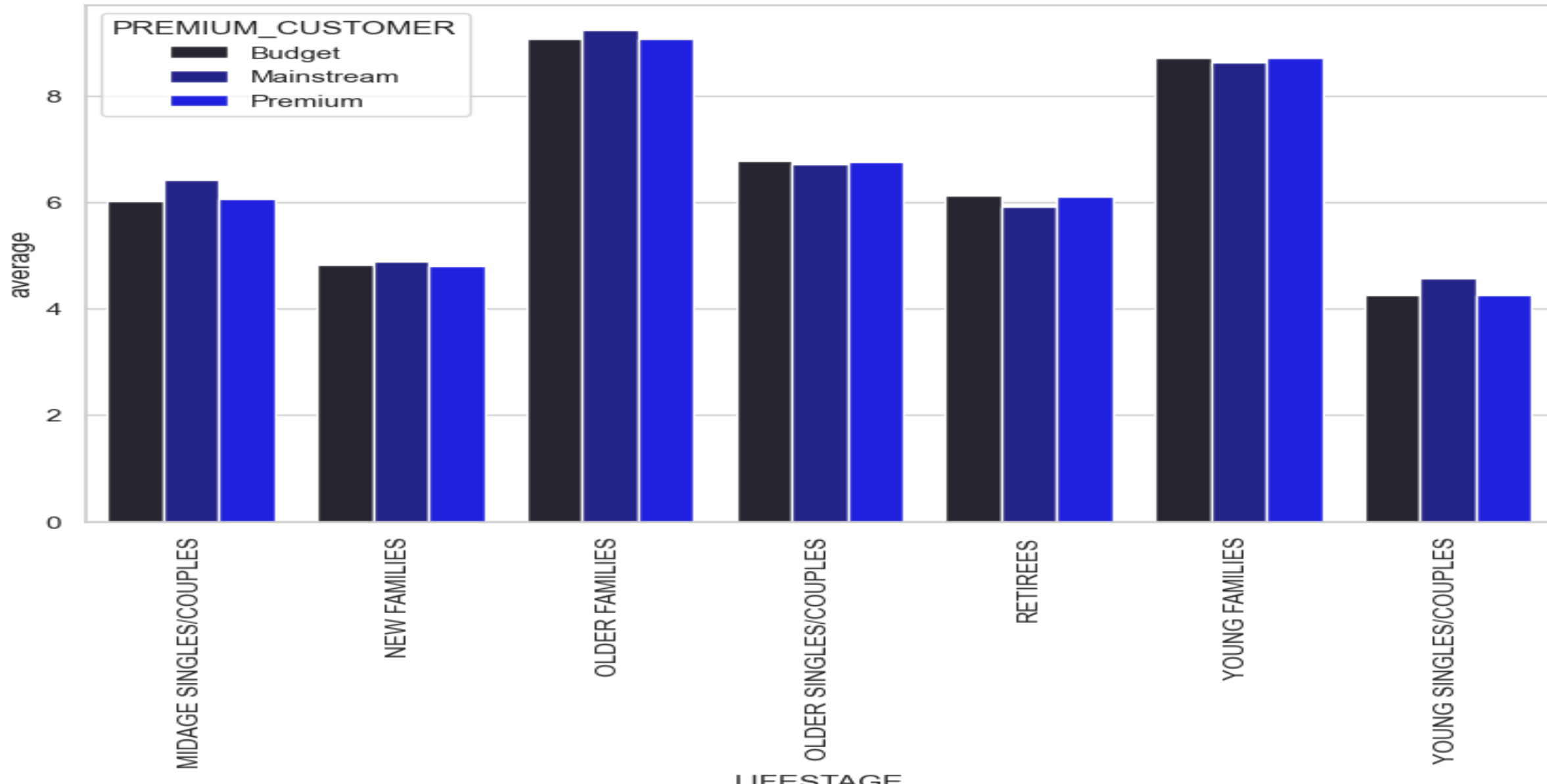
Understand customer purchasing behavior by cleaning and analyzing transaction data.

The number of Chips transitions has remained relatively consistent over the last few weeks; a notable increase occurred in the week before Christmas

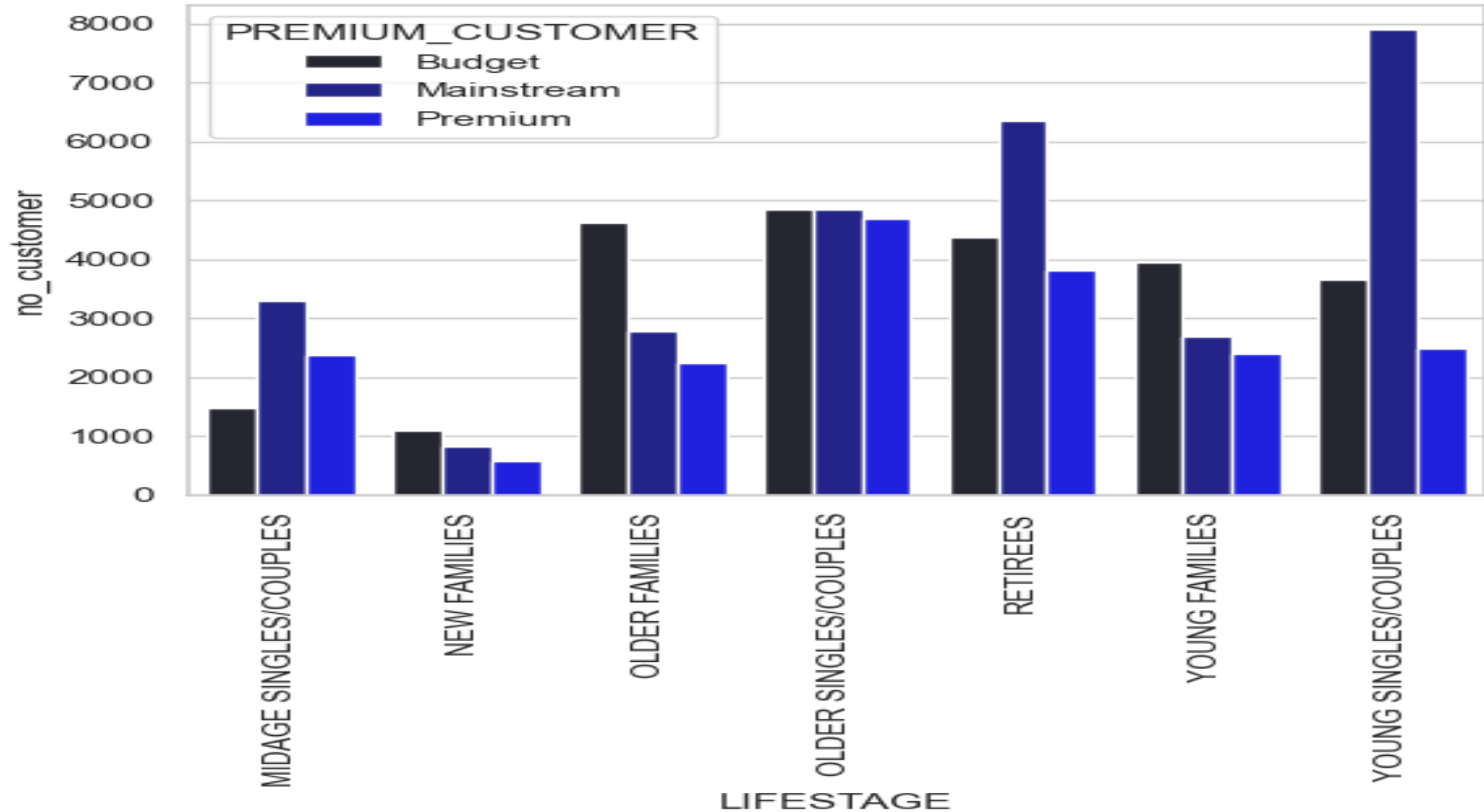


The total number of transactions in the week including Christmas was negatively affected by public holiday store closures

Affluence appears consistent across each individual life stage profile; Older and Young Family shoppers purchase the highest avg units per transaction



Mainstream Young Singles & Couples make up the largest proportion of Snacking Chips shoppers; Mainstream Retirees also have a significant share



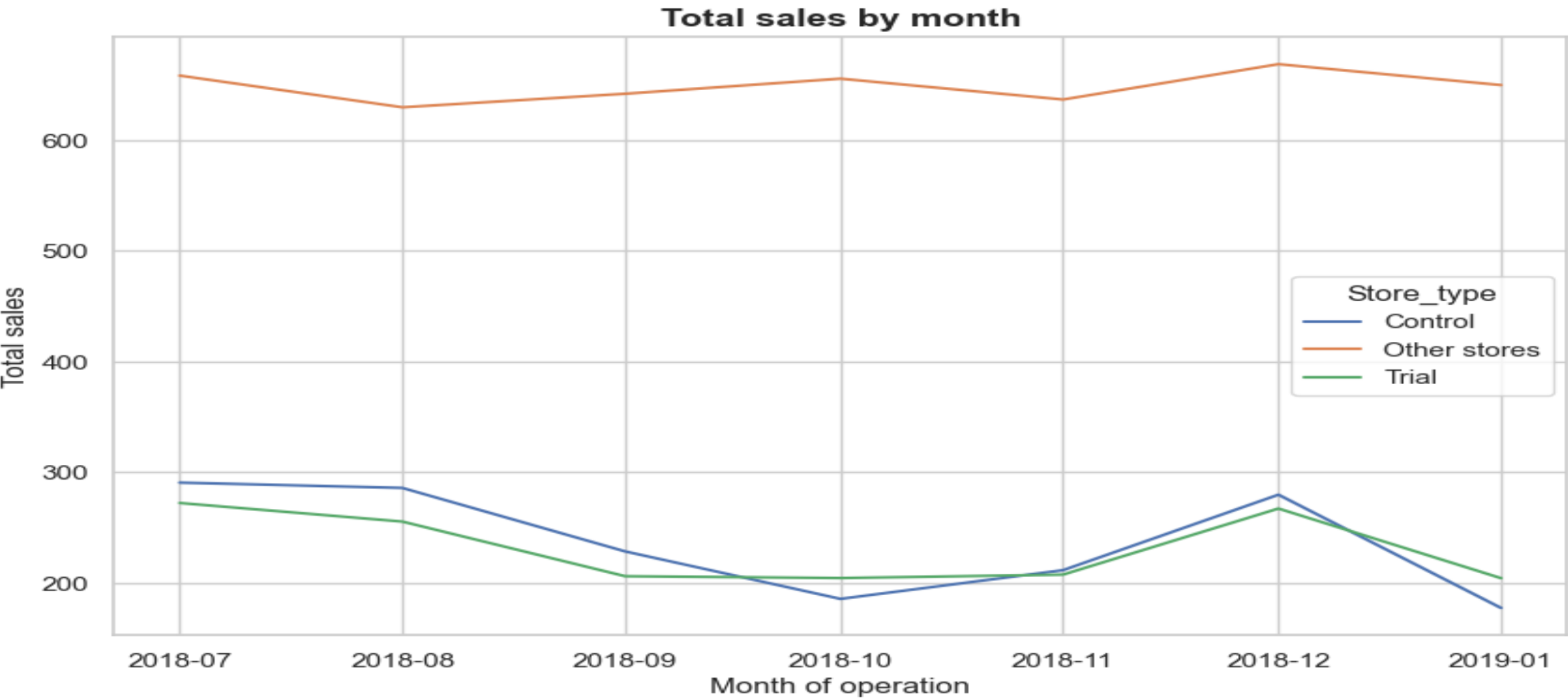
02

Trial store analysis

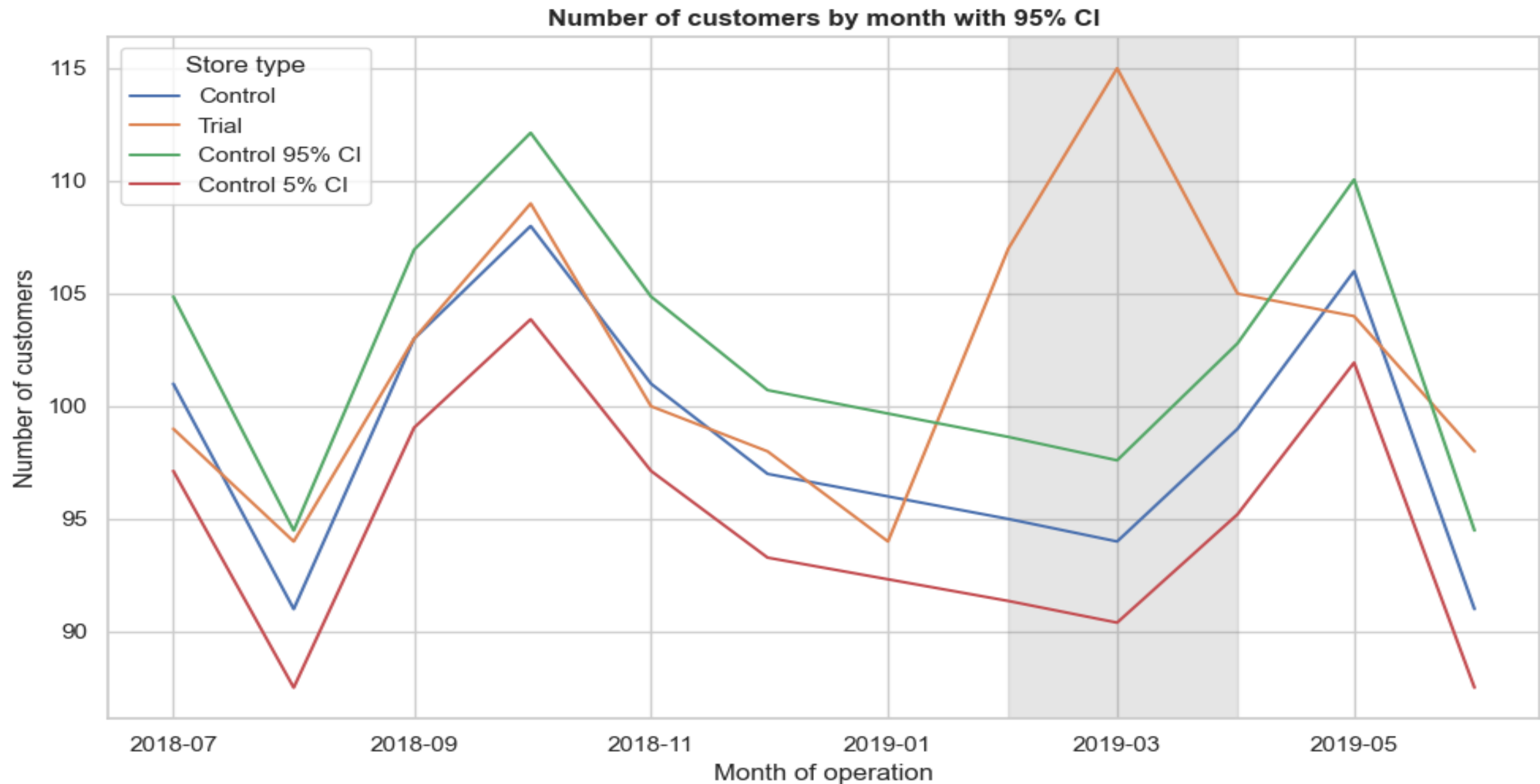
- Analyze trial stores (77, 86, 88) against matched control stores during Feb–Apr 2019.
- Measure uplift in **sales** and **customer counts** to assess the impact of an intervention.
- Use statistical tests and visualizations to validate significance.

Control stores are chosen to be “lookalikes” of trial stores, so the comparison is fair and meaningful

(Avg monthly store sales over time)



From Feb to May the trial store outperformed the control store highlighting the success of the new store layout



Store 77

Sales uplift: Significant in March (+36.7%) and April (+62.3%).

Customer uplift: Also significant in March and April.  **Successful** – strong positive trial effect.

Store 86

Sales uplift: Only significant in March (+31.6%).

Customer uplift: Significant across all three months (Feb–Apr).  **Mixed success** – customer traffic improved, but sales impact was weaker.

Store 88

Sales uplift: Significant in March and April.

Customer uplift: Strong alignment pre-trial, with clear uplift in trial months.  **Successful** – consistent positive effect

Overall Conclusion

- Two out of three trial stores (77 & 88) showed clear, statistically significant increases in both sales and customers.
- **Store 86** showed customer growth but limited sales uplift, suggesting possible differences in implementation.
- **Overall trial outcome:** The intervention was **successful**, demonstrating measurable uplift in customer traffic and sales.

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